by Ivan Aidun

Semidirect products Let N, Q be groups such that Q acts on N on the left. Since N is a group, it seems reasonable to ask about those actions that "respect the group structure". By that I mean that the action satisfies $(q \cdot n_1)(q \cdot n_2) = q \cdot (n_1 n_2)$, where $q \cdot n$ indicates the action of q on the element n. (Another way to say this: instead of thinking of a group action as a homomorphism $Q \to S_N$, we think about it as a homomorphism $Q \to \mathrm{Aut}(N)$.)

If we have such an action, we can construct the semidirect product $N \rtimes Q$. The semidirect product is a group whose underlying set is $N \times Q$, with multiplication (somewhat cryptically) defined by $(n_1, q_1)(n_2, q_2) = (n_1(q_1 \cdot n_2), q_1 q_2)$. You should think of this as *almost* being the direct product $N \times Q$, but the multiplication law has been "twisted" by the action of Q. It might be better to call this *one* semi-direct product of N and Q, because we can get non-isomorphic products by choosing different actions of Q on N (always satisfying $(q \cdot n_1)(q \cdot n_2) = q \cdot (n_1 n_2)$).

With this definition, one can see that $N \rtimes Q$ has a normal subgroup isomorphic to N consisting of elements of the form (n, e_Q), and a subgroup isomorphic to Q consisting of elements of the form (e_N, q). The explanation for the cryptic multiplication is that it is defined so that conjugating an element of the normal subgroup (n, e_Q) by (e_N, q) corresponds to the action of q on n: $(e_N, q)(n, e_Q)(e_N, q^{-1}) = (q \cdot n, e_Q)$. Thus, our original group action has been "encoded" in the semidirect product as an action by conjugation.

The subgroup isomorphic to Q need not be normal, as fixing an element $n \in N$, the set of elements $\{(n(q \cdot n^{-1}), q) : q \in Q\}$ forms a conjugate subgroup.

The semidirect product test Just like with direct products, there's a standard test to show that G is the (internal) semidirect product of its subgroups N and Q. If:

- N is normal in G,
- $N \cap Q = \{e\}$ is the set containing only the identity element, and
- $G = NQ$ (that is, every element $g \in G$ can be written as nq for some $n \in N$ and $q \in Q$)

then $G = N \rtimes Q$ (where the action of Q on N is conjugation in G).

The commutator (derived) subgroup An important subgroup of a group is the commutator subgroup, sometimes also called the derived subgroup. (I will not call it that because I think this terminology is misleading if you're familiar with other uses of the term "derived".) The commutator subgroup of G is denoted $[G, G]$, and it is the group generated by elements of the form $ghg^{-1}h^{-1}$. This means a general element of $[G, G]$ will look like $g_1 h_1 g_1^{-1} h_1^{-1} \dots g_n h_n g_n^{-1} h_n^{-1}$.

The commutator subgroup will be trivial if and only if G is abelian. For any group G, $[G, G]$ is a normal subgroup of G. A tricky way to see this is to suppose $x \in [G, G]$, and $g \in G$, then $gxg^{-1}x^{-1} \in [G,G]$ because it is a commutator, so call $y = gxg^{-1}x^{-1}$. Then we have $gxg^{-1} = yx$, and $yx \in [G,G]$ since both of $x, y \in [G,G]$. (The idea here is that commutators are *almost* like conjugation.)

Furthermore, $G/[G,G]$ is always an abelian group, since in the quotient every commutator is trivial. The group $G/[G,G]$ is called the *abelianization* of G, and you should think of it as the "best abelian approximation" to G. More precisely, if A is an abelian group, and $f\colon G \to A$ is a homomorphism, then there exists a homomorphism $\tilde{f}\colon G/[G,G] \to A$ so that $f = \tilde{f} \circ \pi$, where $\pi\colon G \to G/[G,G]$ is the quotient map. In diagram form:

$$\begin{tikzcd} G \arrow[r, "f"] \arrow[d, "\pi"'] & A \\ G/[G,G] \arrow[ur, dashed, "\tilde{f}"'] & \end{tikzcd}$$

You will also (in the next paragraph) see the notation $[H_1, H_2]$ for H_1, H_2 subgroups of a given group G. In this case, the notation means the subgroup generated by commutators $h_1h_2h_1^{-1}h_2^{-1}$ where $h_i \in H_i$, $i = 1, 2$.

Nilpotent groups A nilpotent group is a generalization of an abelian group. There are two equivalent definitions of nilpotent groups:

- G is nilpotent if the descending sequence of subgroups $G = G_0 \rhd G_1 \rhd G_2 \rhd \dots$ eventually terminates in the trivial subgroup, where $G_i = [G_{i-1}, G]$.
- G is nilpotent if the ascending sequence of subgroups $\{e\} = Z_0 \lhd Z_1 \lhd Z_2 \lhd \dots$ eventually terminates in the full group G, where Z_{i+1} is defined so that $Z_{i+1}/Z_i = Z(G/Z_i)$. (Confirm that this gives a well-defined subgroup!)

These are respectively called the **lower central series** and the **upper central series**. If these series terminate, their lengths are the same, and the length of either series is called the *nilpotency class* of G. The groups of nilpotency class 1 are exactly the abelian groups.

A sometimes useful fact is that a group is nilpotent if and only if it is the (internal) direct product of its Sylow-p subgroups.

Solvable groups Another generalization of abelian groups are solvable groups. A group is solvable if there exists a sequence of subgroups $G = G_0 > G_1 > \cdots > G_n = \{e\}$ such that:

- each G_i is a normal subgroup of G_{i-1}, and
- the quotients G_{i-1}/G_i are all abelian.

The terminology comes from Galois theory, as a polynomial can be solved by radicals if and only if its Galois group is a solvable group.

Given a group, it might be difficult to come up with such a sequence of subgroups just by thinking and messing around. Luckily, there's a standard sequence that you can always check.

Given a group G, the **derived series** of G is the sequence of subgroups $G \triangleright G^{(1)} \triangleright G^{(2)} \triangleright \dots$ where $G^{(i)}$ is the commutator subgroup of the preceding group $[G^{(i-1)}, G^{(i-1)}]$. (I would prefer to call this the "commutator series", but it seems like in actual practice nobody does that.) **Note:** the derived series is usually not the same as the lower central series; the difference is $G^{(i)} = [G^{(i-1)}, G^{(i-1)}]$ versus $G_i = [G, G_{i-1}]$.

It's not difficult to check that a finite group is solvable if and only if its derived series eventually reaches the trivial group. (One direction is obvious, the other direction you can induct on $|G|$.)